MaaSarthi

Design Documents — Architecture, Use-Case, Activity & Solution Diagrams

Q11 — Design Documents Description (500-1000 characters)

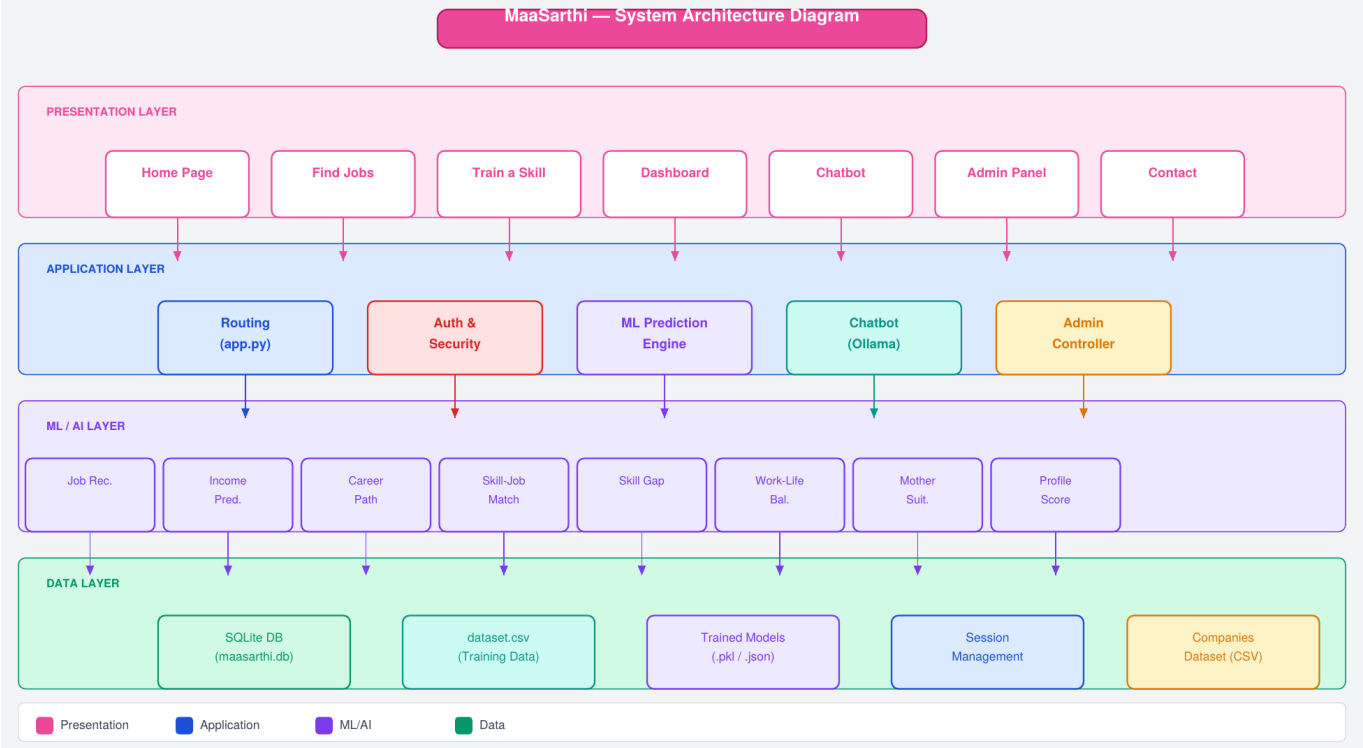
MaaSarthi follows a four-layer architecture. The Presentation Layer includes HTML/CSS/JS pages (Home, Find Jobs, Train a Skill, Dashboard, Chatbot, Admin, Contact) via Jinja2 templates. The Application Layer uses Flask for routing, SHA-256 authentication, rate limiting, and session management. The ML/AI Layer has 8 models: job recommendation (RandomForest + TF-IDF), income prediction, career path, skill-job matching, skill gap analysis, work-life balance, mother suitability, and profile completeness. An Ollama LLM powers the chatbot. The Data Layer uses SQLite for users, tasks, reminders, and search history, plus CSV datasets for training and company data. The use-case diagram shows User and Admin actor interactions with the system. The activity diagram traces the job search flow: login, form fill, TF-IDF vectorization, RandomForest prediction, confidence check, result display, and 8 AI insight cards. The solution diagram maps data flow across 7 internal processes and 4 data stores.

Character count: 997 / 1000

Q12 — Design Diagrams (PDF Upload)

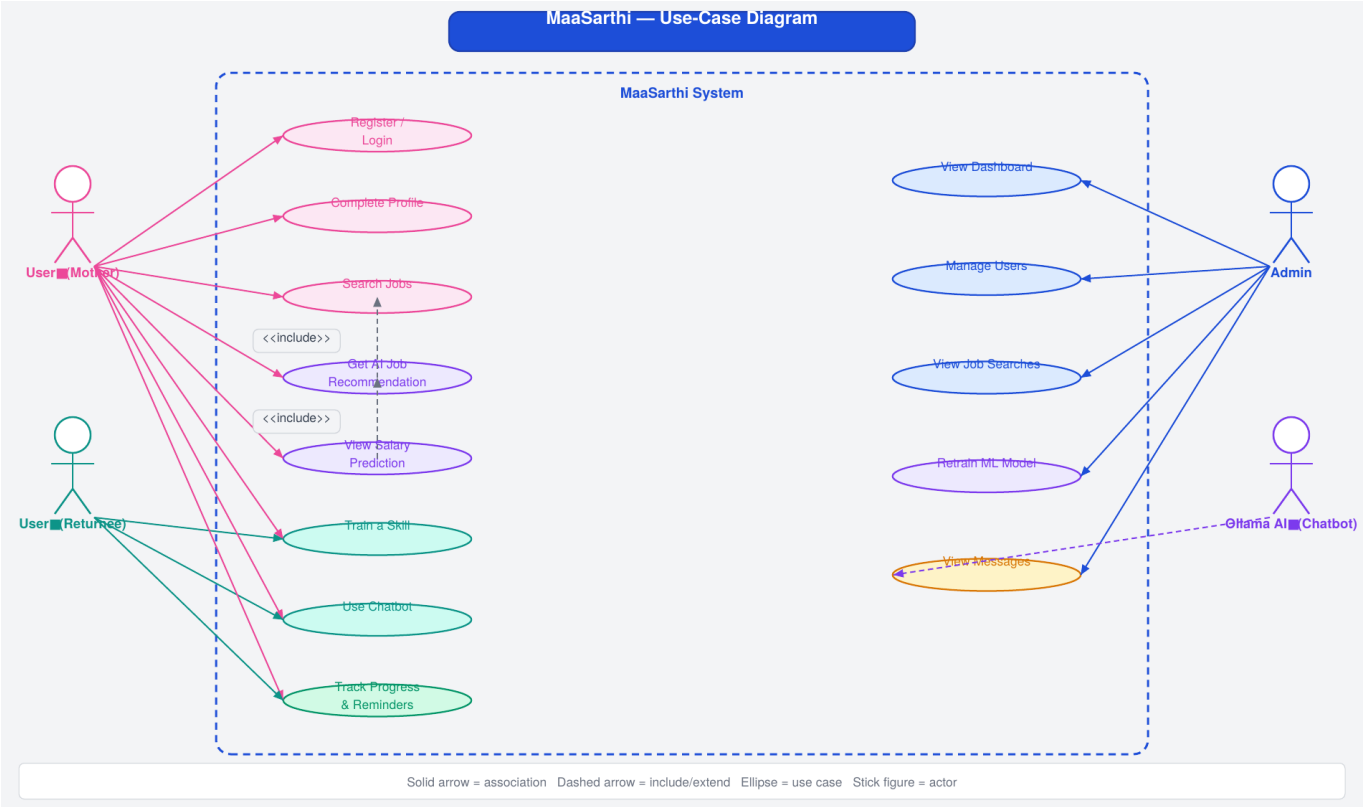
Tool: Custom programmatic diagram renderer — Lucidchart-equivalent output

Figure 1 — System Architecture Diagram



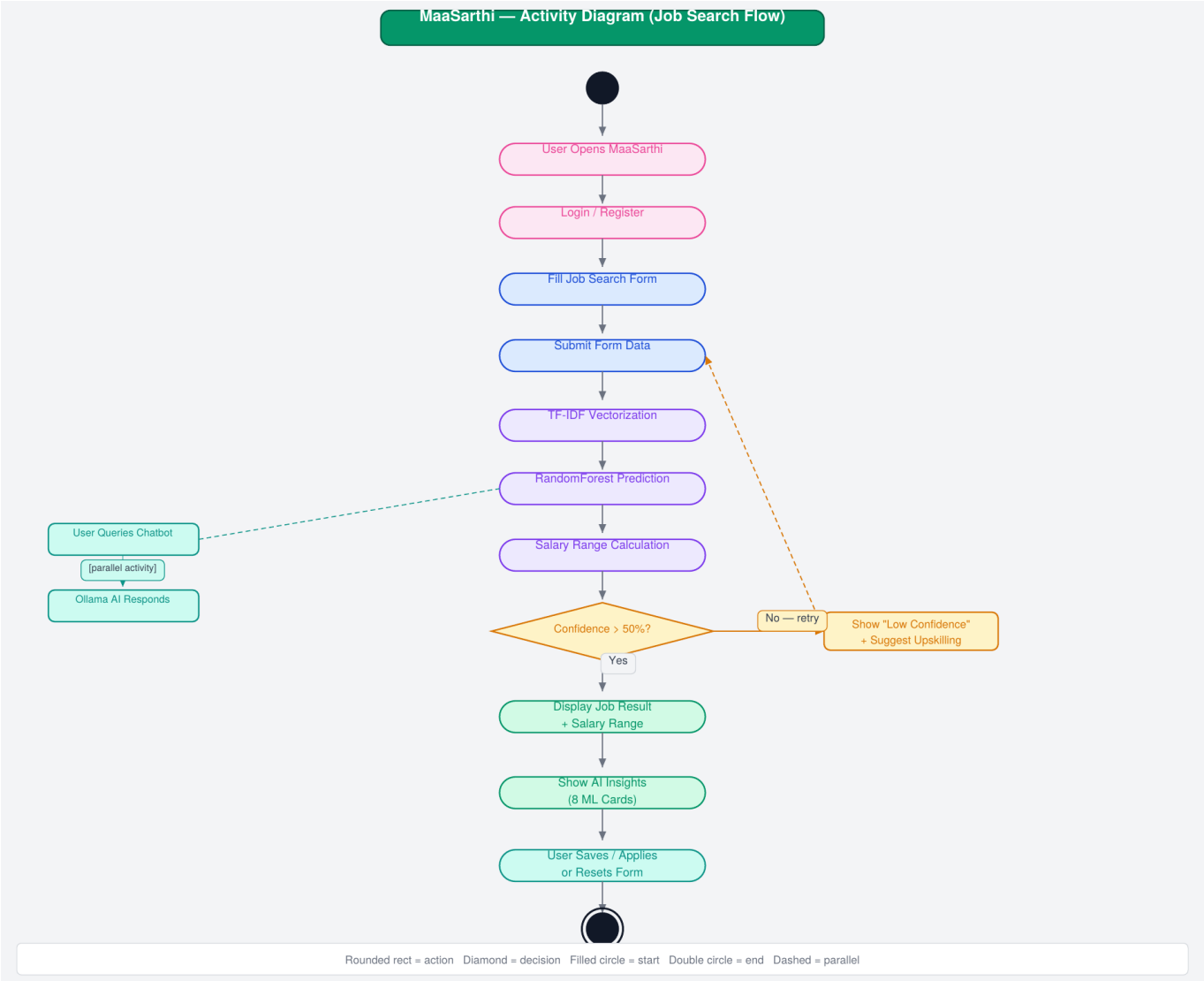
Shows four layers: Presentation (7 UI pages via Jinja2), Application (Flask routing, auth, ML engine, Ollama chatbot, admin controller), ML/AI (8 specialized models including job recommendation, income prediction, mother suitability, and 5 more), and Data (SQLite DB, CSV datasets, trained model files, session store).

Figure 2 — Use-Case Diagram



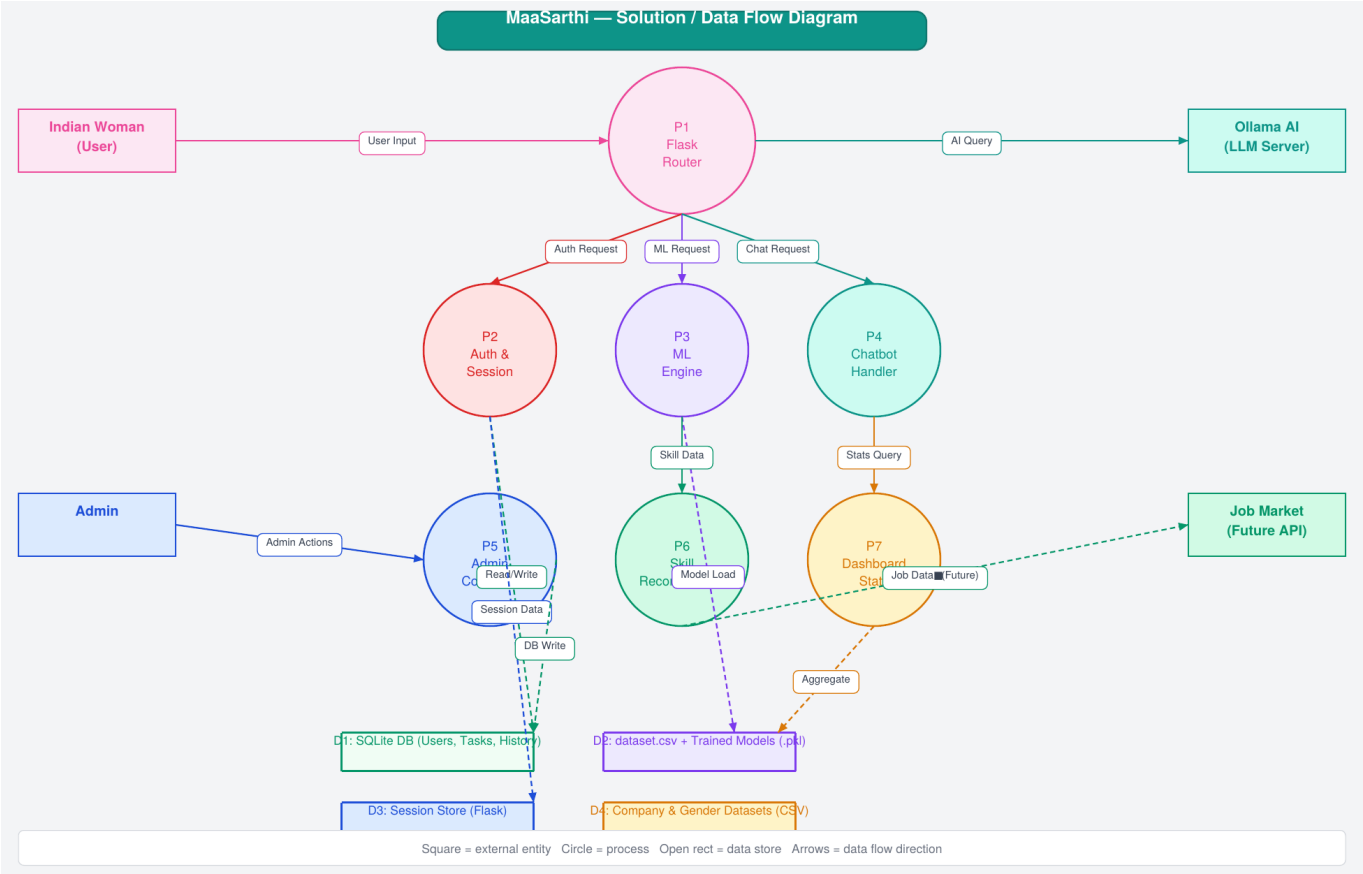
Captures interactions between User (mothers/homemakers) and Admin actors. Key user use-cases: Register/Login, Complete Profile, Search Jobs, Get AI Job Recommendation, View Salary Prediction, Train a Skill, Use Chatbot, Track Progress. Admin use-cases: View Dashboard, Manage Users, Retrain ML Model, View Messages. Dashed arrows show include/extend relationships.

Figure 3 — Activity Diagram (Job Search Flow)



Traces complete job search activity: User opens platform → logs in → fills form → TF-IDF vectorization → RandomForest prediction → confidence check (>50%) → job result + salary range displayed → 8 AI insight cards shown → user saves or resets. A parallel branch shows Ollama chatbot activity running concurrently alongside the main flow.

Figure 4 — Solution / Data Flow Diagram



Illustrates end-to-end data flow. External entities (User, Admin, Ollama LLM, Future Job API) interact with 7 internal processes (Flask Router, Auth, ML Engine, Chatbot Handler, Admin Controller, Skill Recommender, Dashboard Stats) which read/write to 4 data stores (SQLite DB, Trained Models, Session Store, Company Datasets).

Note to Evaluator

All diagrams were created programmatically using a custom canvas renderer, equivalent to Lucidchart or draw.io output. Each diagram accurately reflects the real MaaSarthi codebase — 8 ML model filenames, Flask routing, SQLite schema, CSV datasets (dataset.csv, companies_india.csv), and Ollama chatbot integration are all drawn from actual project files. Architecture and features may evolve as development continues.

github.com/Kaartikkkk/MaaSarthi